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A Comparative Investigation on the CO_2 Sequestration Potential in a Heavy Oil Reservoir and Low Permeability Reservoir

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Abstract

CO_2 injection into reservoirs for enhanced oil recovery and sequestration is the most efficiency way to realize the carbon neutrality goal. Various types of reservoirs could be the targets for CO_2 sequestration but which one should be the best choice is still unclear, especially for the huge reserves of heavy oil reservoirs and low permeability reservoirs. The sealing property of cap rock, displacement efficiency of CO_2 flooding and CO_2 dissolution ability in the oil jointly determined the CO_2 sequestration potential.

This work compared the CO_2 sequestration potential between heavy oil reservoirs and low permeability light oil reservoirs via a series of experiments and numerical simulations. The breakthrough pressure of CO_2 in the caprock, the CO_2 dissolution amount in heavy oil and light oil and oil displacement effective by CO_2 flooding were investigated to evaluate CO_2 sequestration capability via the dissolution and capillary trapping in two types reservoirs. Moreover, the numerical simulation of CO_2 flooding in a heavy oil reservoir and low permeability reservoir were performed based on the above experimental data to quantitatively clarify the CO_2 sequestration potential.

The breakthrough pressure of the mudstone caprock of the low permeability reservoir was 3.5 times higher than that of the mudstone caprock of the heavy oil reservoir, which indicated the CO_2 sequestration pressure in the low permeability reservoir was higher than that in the heavy oi reservoir. The solubility of CO_2 in the heavy oil was 2 times lower than that in the light oil. The CO_2 sequestration amount per unit pore volume in the low permeability light oil reservoir was slightly higher than that in the heavy oil reservoir under a certain temperature and pressure condition, and it increased while temperature increased due to the incremental displacement efficiency of heavy oil. The proportion of CO_2 sequestration by capillary trapping in low permeability cores could reach 87% while the CO_2 sequestration by dissolution in the heavy oil reservoir was nearly 30%. Numerical simulation found that in the real reservoir conduction, the residual trapping proportion in the low permeability reservoir was lower than that in the heavy oil reservoir

but the effective sequestration coefficient of low permeability reservoir was larger than that of heavy oil reservoir. We expect this work could point out the application direction for the CO_2 enhanced oil recovery and sequestration.

Introduction

CO_2 injection into the reservoir for oil and gas recovery enhancement and then sequestration is the most efficiency way to realize the carbon neutrality goal (Edouard et al. 2023; Liu et al. 2025; Song et al. 2025). CO_2 featured with high solubility with oil. It can swell crude oil, reduce oil viscosity and extract the light components of crude oil, which make it become a good candidate for enhance oil recovery (EOR) (Liu et al. 2022; Sambo et al. 2023; Wang et al. 2025). Many oilfields around the world have developed CO_2-EOR and sequestration projects (Qin et al. 2015; Zhang et al.; Sacuta et al.2017; Gunnarsson et al. 2018). It includes low permeability reservoirs and heavy oil reservoirs. It is still unclear which one could be the target for CO_2-EOR and sequestration.

Low permeability reservoirs characterized by the unfavorable porosity permeability relationships and accompanied by a large amount of clay minerals, making it is difficult for the water injection (Zhang et al. 2025). Many works had verified CO_2 flooding could achieve a remarkable oil recovery in the low permeability reservoirs (Zhou et al.2019). Fakher and Imqam (2020) investigated the impact of CO_2 injection pressure on the oil recovery in low-permeability shale reservoirs and found the CO_2 injection pressure increment could enhance CO_2 adsorption capacity. He believed CO_2 flooding could reach the optimal displacement efficiency when the injection pressure reached or exceeded the minimum miscibility pressure (MMP). Cai et al. (2021) also found the displacement efficiency of CO_2 miscible flooding could be twice more than that of the immiscible flooding. Therefore, the relationship between the reservoir pressure and CO_2 MMP is very important. Since the original reservoir pressure of the low permeability reservoir is usually high, CO_2 is easy to reach the MMP with crude oil. CO_2 becomes the preferred displacement media for the reservoir whose condition could meet the requirement of CO_2 immiscible flooding. It should be noted that MMP is not necessary for all the reservoirs. Cronin et al. (2018) believed MMP was not significant for CO_2 flooding in the ultralight reservoirs because the CO_2 miscible flooding relied on the advection-dominated mass transfer but diffusion dominated the ultra-tight reservoirs development.

In addition to the application of CO_2 in the low permeability light oil reservoir, heavy oil reservoir is another target for its application. Many experiments and numerical simulations have been performed to investigate the possibility for CO_2 assistant heavy oil thermal recovery (Derakhshanfar et al. 2012; Wang et al. 2018). Miller and Jones (1981) found CO_2 dissolved into the heavy oils could reduce their viscosity by using three heavy oils with viscosities from 800 to 4000 mPa·s. Sohrabi et al. (2007) compared the heavy oil recoveries among CO_2 flooding, water flooding and CO_2 assistant steam flooding. They found that CO_2 assistant steam flooding could increase oil recovery by 30% over the water flooding. Liu et al. (2016) proposed combing viscosity reducer, CO_2 and steam could achieve a viscosity reduction ratio of 84.38% even the viscosity of heavy oil was more than 12×10^4mPa·s. Shi et al. (2019) found the oil recovery of CO_2-assisted hot-water flooding at 120°C could be comparable to that of 200 °C steam flooding because of the significantly viscosity reduction of the heavy oil after CO_2 dissolved. It can be known CO_2 was a good assistant agent for heavy oil recovery. But there was no reported whether it could be used alone for those less viscous oil recoveries.

After CO2-EOR in the reservoirs, its sequestration should be paid much attention. Structural trapping, dissolution trapping and mineral trapping are three main ways for CO_2 sequestration. Structural trapping provided the primary contribution during the early-stage of CO_2 sequestration, whereas other mechanisms gained importance over time as the sequestration stability improved. Mineral trapping was the final phase and often initiated at the millennia post-injection (Kim et al., 2023). However, the proportion of these sequestration methods in different reservoirs are still unclear. The work is to compare CO_2 enhanced oil recovery and sequestration potential in a heavy oil reservoir and low permeability reservoir via the experimental research and numerical simulation.

Experiments and Simulation

Experimental Materials and Instruments

Two crude oils used in the experiments were from two pilots in LiaoHe Oilfield. One is light oil with a viscosity of 3.5 mPa.s under the reservoir temperature of 85°C, while other is heavy oil with the viscosity of 369 mPa.s at the current reservoir temperature of 62°C after multiple rounds of steam huff and puff. The cap rock of the low permeability reservoir was mudstone while that of the heavy oil reservoir was argillaceous siltstone. They were acquired for the breakthrough pressure measurement and their permeabilities were both lower than 0.1mD. The reservoir cores from the low permeability reservoir with a permeability of 3mD and that from the heavy oil reservoir with a permeability of 452mD were used for the core flooding experiments. The ion composition of formation water was shown in Table 1. The gas used in the experiments is CO_2 with a purity of 99.9%.

Table 1—Composition of the formation water

Reservoir type	Na^+/K^+	Ca^{2+}	Mg^{2+}	Cl^-	HCO_3^-	CO_3^{2-}	TDS (mg/L)
Heavy oil reservoir	1014.3	8.02	/	159.6	1586.5	420	3188.4
low permeability reservoir	1707.8	15.5	24.7	758	2929	/	5435

Breakthrough Pressure of Cap Rock Measurement

The breakthrough pressure measurement was followed Petroleum and Natural Gas Industry Standard of China (SY/T 5748-2020). The core was fully saturated the formation water and placed into the core holder. The confining pressure was constant and equal to the overburden pressure of the reservoir. The initial injection pressure difference was set at 2 MPa when the core permeability was between 0.01mD to 0.1mD and 5 MPa when the core permeability was between 0.001mD to 0.01mD. Then gradually increasing the pressure till bubbles uniformly and continuously escaped from the outlet of the core holder. The corresponding pressure was the gas breakthrough pressure. The stable time and pressure difference interval of each stage could refer to the standard (SY/T 5748-2020).

CO_2 Solubility in Crude Oil Measurement

The equilibrium liquid sampling analysis method was used for the CO_2 solubility measurement. CO_2 was injected into a certain volume of solution in the container under some temperature and pressure. After sufficient dissolution, the crude oil dissolved with CO_2 was released via a pump pushing the piston container. The volume of released oil and CO_2 were recorded and then the solubility of CO_2 could be calculated. The experimental procedure was shown in Fig. 1. The effect of pressure and temperature on CO_2 solubility in two crude oils were investigated in this section. Comprehensively consider the actual reservoir and the coverage range of the data, the pressure from 10MPa to 25MPa was selected for heavy oil experiments while it was from 15MPa to 35 MPa for light oil experiments. The temperature from 30°C to 70°C was selected for heavy oil experiments while it was from 50°C to 110°C for the light oil experiments.

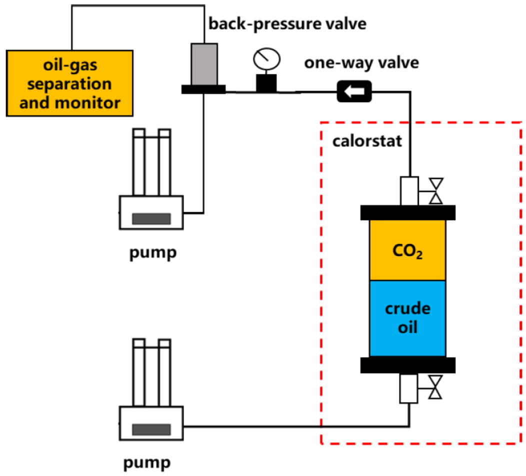

Figure 1—Schematic of the experiment for CO_2 solubility measurement

NMR Monitoring Pore-scale Oil Mobilization and CO_2 Sequestration

The CO_2 displacement experiments and NMR scanning were performed to understand the oil mobilization efficiency and CO_2 sequestration state in the heavy oil reservoir and low permeability reservoir. The cores were vacuumed and saturated with a $MnCl_2$ solution with the concentration of 2000 mg/L to shield the NMR signal in the aqueous. Two kinds of crude oils were separately injected into the corresponding cores to create initial oil saturations. The flow rate increased for oil fully saturated when the oil was breakthrough. The initial oil saturation was calculated and NMR T_2 spectrums were scanned to obtain the initial state. Then set the back pressure based on the reservoir pressure and continue injecting oil for the pressure increment until reaching the experimental pressure. The initial pressure of low permeability reservoir was 35 MPa and that of heavy oil reservoir was 10 MPa. These two pressures were separately set as the back pressure in the core flooding. The displacement velocity of CO_2 was set at 0.05mL/min. The displacement pressure difference and oil production were recorded. The NMR T_2 spectrum was scanned after CO_2 flooding to clarify the remaining oil distribution in each pore. Combining with CO_2 solubility in brine and water, the amount proportion of CO_2 residual trapping and dissolution sequestration could be distinguished. The schematic of CO_2 flooding experiment was shown in Fig. 2.

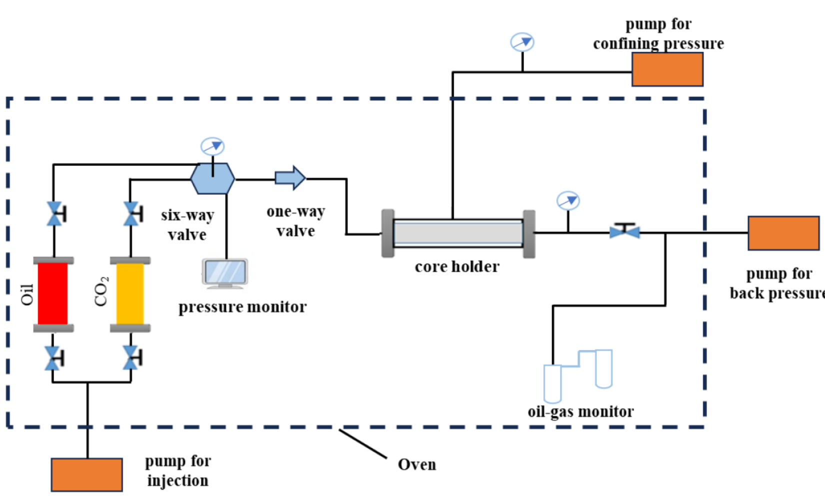

Figure 2—Schematic of CO_2 displacement experiments

The method for the conversion between NMR T_2 spectrum and mercury injection curve could be found in our previous work (Liu et al. 2018). The conversion coefficients of two cores were respectively determined because their permeability difference was large. Fig. 3 presented the relationship between relaxation time T_2 and pores sizes after conversion. Now NMR scanning could be used for the analysis of oil mobilization and CO_2 sequestration in each pore size.

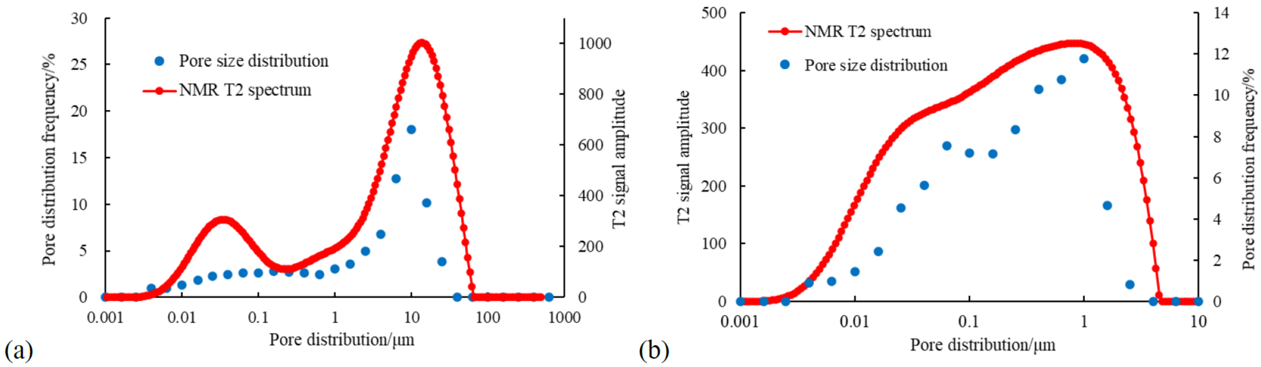

Figure 3—Corresponding relation between relaxation time T_2 and pores sizes after conversion: (a) heavy oil reservoir; (b) low permeability reservoir

Numerical simulation

Numerical simulation was conducted to compare CO_2 enhanced oil recovery and sequestration potential of the heavy oil reservoir and low permeability light oil reservoir. These models were calibrated with the phase behavior data of original formation fluids, along with the geological characteristics and production data of two pilots

Reservoir Fluids Model

Based on chromatographic measurements of degassed oils and associated gases from the pilots, compositional simulations were performed using the WinProp module of phase behavior simulation software. Simulations were conducted at original reservoir pressure and temperature, utilizing a representative gas-oil ratio to configure live oil properties and reconstruct in-situ fluid compositions. Phase equilibrium calculations employed the Peng-Robinson (PR) equation of state and yielded the pseudo-component composition of the model, as shown in Table 2. The model was fitted to experimental data including saturation pressure, viscosity, gas-oil ratio and multi-stage degassing results for the reservoir crude oil. The parameters used in the numerical simulation were summarized in Table 3.

Table 2—Pseudo-component composition of live oil in the model

Light oil		Heavy oil	
component	Molar composition /%	component	Molar composition/%
N_2	0.02	N_2	0.48
CO_2	0.36	CO_2	0.70
CH_4	48.67	CH_4	17.24
C_2-C_5	9.58	C_2-C_4	5.18
C_6-C_{12}	10.46	C_5-C_{10}	21.43
C_{13}-C_{21}	15.07	C_{11}-C_{19}	9.02
C_{22}-C_{29}	12.47	C_{20}-C_{31}	9.77
C_{30}-C_{38}	3.37	C_{32}-C_{36}	36.19

Table 3—Fitting results of fluid phase features

Reservoir type	Saturation pressure/MPa			viscosity/(mPa•s)			Gas-oil ratio/(m^3/m^3)		
	Experimental value	Simulation value	Error	Experimental value	Simulation value	Error	Experimental value	Simulation value	Error
Low permeability	15.6	16.2	3.84%	3.2	3.1	3.20%	91.5	89.2	2.50%
Heavy oil	7.4	7.46	0.81%	539.2	539.3	1.82%	16	16.3	3.13%

Geologic Model and Simulation Scheme

A representative well group was selected based on the geological model of the pilots. The production history matching was performed to enhance the reliability and accuracy of the simulation results. Subsequently, numerical simulations were conducted to investigate CO_2 flooding and sequestration performance. Two numerical simulation models of the heavy oil reservoir and low permeability reservoir were developed. The model of low permeability reservoir comprised 31,050 grids, exhibited an initial average formation pressure of 30.24 MPa, the averaged permeability of 21 mD, the averaged porosity of 16.3%, the initial oil saturation of 0.55 and the rock compressibility of 4.5×10^{-6} kPa^{-1}. The heavy oil reservoir model contained 26,325 grids with an initial average formation pressure at 14.70 MPa, the averaged permeability of 978 mD, the averaged porosity of 25.8%, the initial oil saturation of 0.65 and the rock compressibility of 3.0×10^{-6} kPa^{-1}. The three-dimensional schematic diagrams of the selected well-group models for two type reservoirs were shown in Fig. 4. The current remaining oil distribution of two well-group models were shown in Fig. 5 after the history matching.

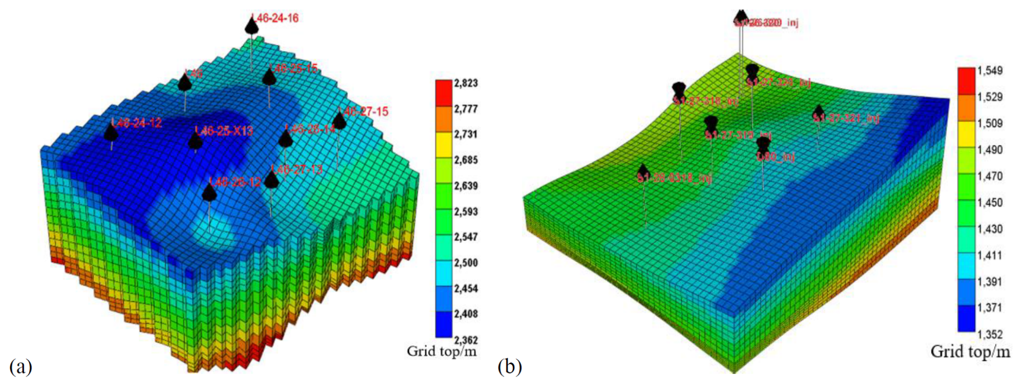

Figure 4—Typical well group model: (a) low permeability reservoir; (b) heavy oil reservoir

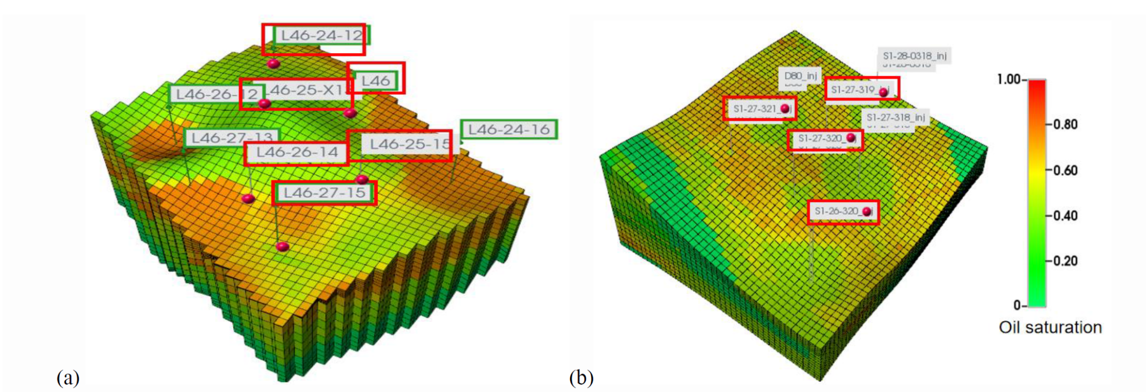

Figure 5—Remaining oil distribution after history matching: (a) low permeability reservoir; (b) heavy oil reservoir

Based on the historical fitting model, wells were arranged according to the current distribution of remaining oil. CO_2 injection was prioritized for wells with low oil saturation in the vicinity, while the high-pressure gas injection was employed to maximize gravity-driven oil displacement. The injection gas wells were highlighted by a red box in Fig. 5, while the remaining wells were designated as the production wells. The gas injection capacity was calculated based on the principle that the bottomhole pressure should not exceed 90% of the reservoir rupture pressure. The constant-pressure production was adopted for the production wells. Their pressures were set based on the reservoir pressure of each well's grid cell after the history matching. During the injection process, the production well was shut down when the production gas-oil ratio reached 2,000 m^3/m^3. And then all wells were switched to CO_2 until the formation pressure equaled the original formation pressure.

Results and Discussion

Breakthrough Pressure of two type reservoirs

Based on the breakthrough pressure test standard, the pressure difference gradually increases till bubbles appeared. The curves in Fig. 6(a) presented the process of CO_2 breakthrough. The zero points in the curve with time increasing corresponded to the gas production rate with the pressure increasing. The pressure at which the gas production occurred was the breakthrough pressure. It was 3.9 MPa for the argillaceous siltstone, which was much lower than that of 13.4 MPa for the mudstone, as shown in Fig. 6(b). The reason was the pores space of silts were larger than those in the mudstone. The permeability of argillaceous siltstone was also larger than that of mudstone. The maximum CO_2 sequestration pressure should be lower than the pressure of the breakthrough pressure added the initial reservoir pressure. The initial reservoir pressure of the heavy oil reservoir was lower than that of the low permeability reservoir. In addition, the breakthrough pressure of the heavy oil reservoir was also low. It could be known the maximum CO_2 sequestration pressure of the heavy oil reservoir must be much lower than that of the low permeability reservoir.

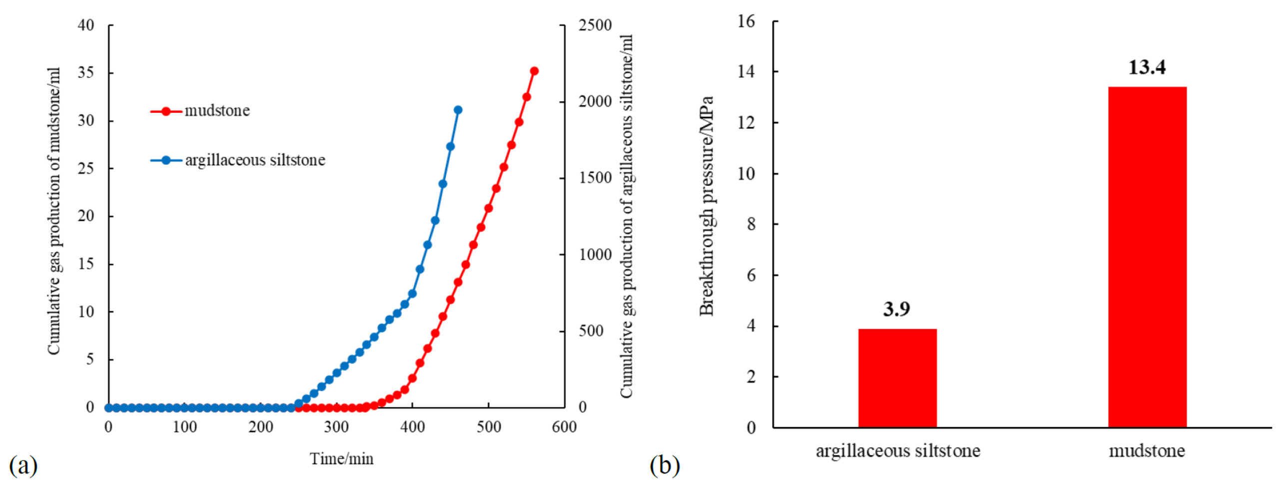

Figure 6—Breakthrough pressure of two kinds of cap rocks measurement results: (a) gas production curves under different pressure differences; (b) breakthrough pressure of two rocks

Comparison of CO_2 solubility in the light oil and heavy oil

The results of CO_2 solubility in the light oil and heavy oil at various temperatures and pressures were shown in Fig. 7. CO_2 solubility was positively correlated with temperature and negatively correlated with pressure. The CO_2 solubility of the light oil was much higher than that of the heavy oil at the same pressure and temperature. It could deduce that CO_2 solubility of the light oil was more sensitivity to the pressure than the heavy oil from the slope of the solubility versus pressure curves. At the current oil reservoir conditions, the CO_2 solubility of the low permeability light oil reservoir was 120 m^3/m^3, which was three times than the heavy oil reservoirs. The low permeability light oil reservoir presented a better solubility sequestration potential.

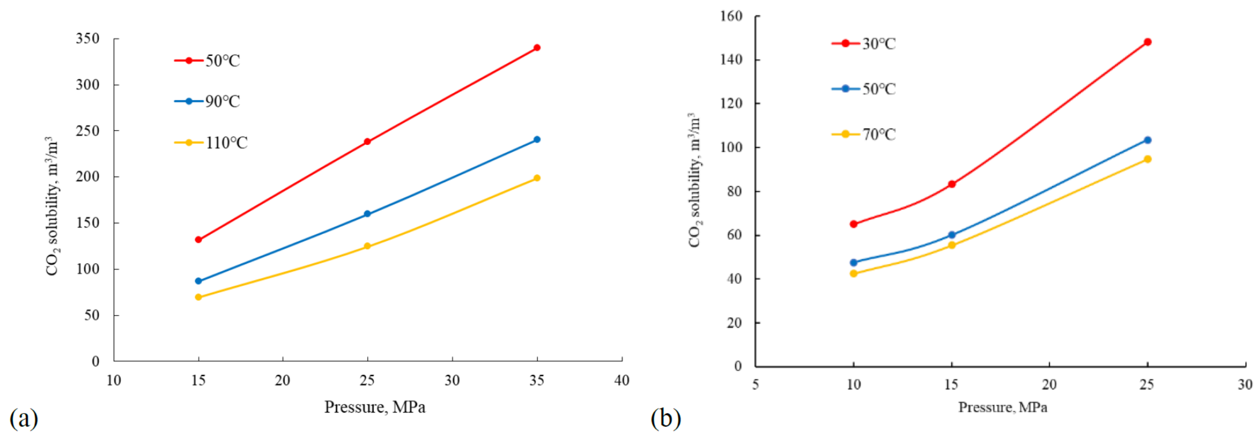

Figure 7—Effect of pressure and temperature on CO_2 solubility: (a) light oil; (b)heavy oil

Quantitative evaluation of CO_2 sequestration state at the pore-scale

Two reservoir cores were separately conducted NMR scanning before and after CO_2 flooding. The changing of T_2 signal amplitude could reflect oil production amount. The oil mobilization efficiency of each pore size could be known since T_2 relaxation time had been converted into the pore size distribution. The oil recovery factor of CO_2 flooding in the low permeability core was 88.6% via the NRM spectroscopy in Fig. 8(a) calculation. It was only 38.9% in the heavy oil core which was calculated using the NRM spectroscopy in Fig. 8(b). The experimental pressure for the low permeability light oil was 35MPa. CO_2 could realize miscible with oil at this pressure, which benefited to a very high oil recovery factor. The viscosity of heavy oil could not significantly decease during the displacement process due to the influence of the non-equilibrium mass transfer between the heavy oil and CO_2. There were lots of residual oil after CO_2 flooding.

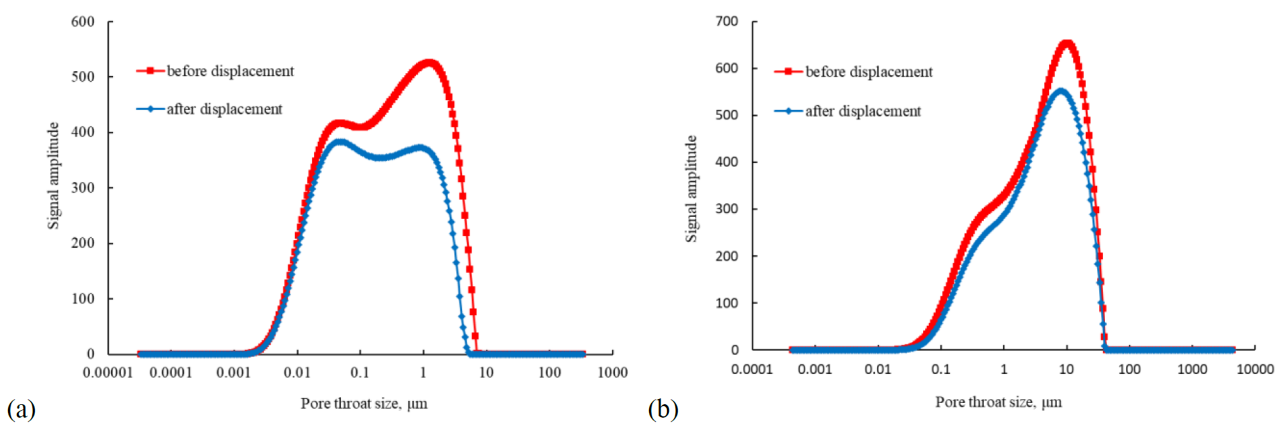

Figure 8—NMR spectroscopy before and after CO_2 displacements: (a) low permeability core with light oil; (b) high permeability core with heavy oil

The pore sizes lower than 0.01μm were defined as the micropore for both heavy oil reservoir and low permeability reservoir. The pore sizes larger than 10 μm were defined as the macropore for the heavy oil reservoir while it larger than 1μm was defined as the macropore for the low permeability reservoir. The others were mesoporous. The contribution of each pore on total oil recovery was calculated using the NMR spectroscopy in Fig. 8. Fig. 9 indicated the macropore contributed much for oil recovery in the low permeability reservoir while the mesoporous dominant oil recovery in the heavy oil reservoir. The residual trapping efficiency equaled to the mount of CO_2 trapped in pores divided the total injected CO_2 amount. The experiments finished when there was no oil coming out from the outlet of cores. The finish time of the heavy oil core flooding experiment should be earlier than that of the low permeability core flooding experiment. But the injection amounts were both set at 1.5PV for the comparison. The residual trapping efficiency of CO_2 flooding in the heavy oil core was 18.6%, which was much lower than that of 40.1% in the low permeability core. It indicated lots of CO_2 would be escaped from the production well in the heavy oil reservoir.

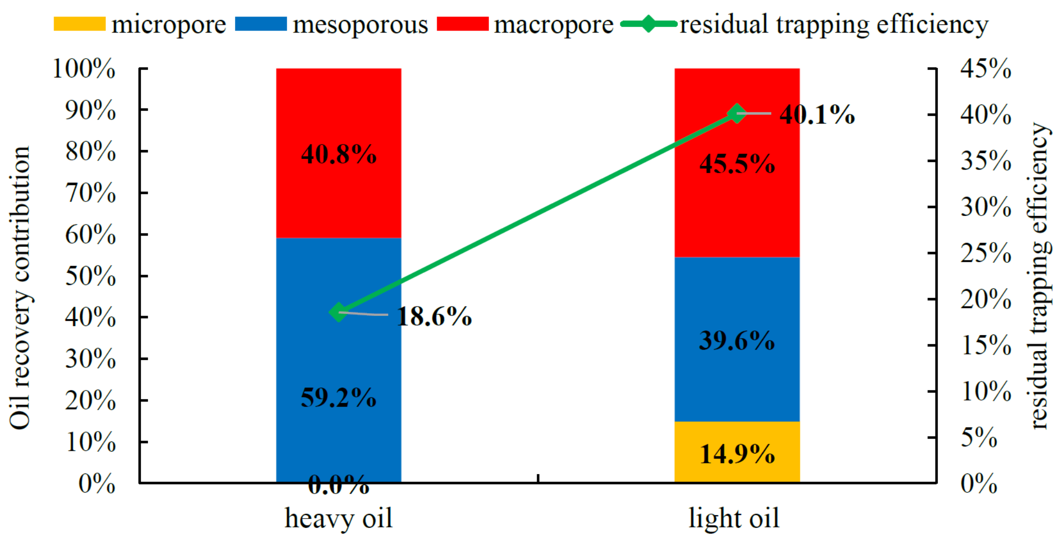

Figure 9—Contribution of each pore on oil recovery and CO_2 residual trapping efficiency of two oils

Three CO_2 sequestration states including residual trapping, CO_2 dissolution in oil and CO_2 dissolution in water could be distinguished via the oil displacement efficiency and CO_2 solubility data. The CO_2 solubility in two oils could be found from Fig. 7. Since the salinity of two formation water was almost similar, the CO_2 solubility in brine could be found from reference (Duan et al. 2003). The calculated results in Fig. 10(a) indicated residual trapping was the main sequestration state in both heavy oil core and low permeability core. Since the CO_2 displacement efficiency in the low permeability core was much higher than that in the heavy oil core, there were more space for CO_2 sequestration. The residual trapping proportion in the low permeability core was higher than that in the heavy oil core. The CO_2 dissolution in the heavy oil core was higher than that in the low permeability core because there were more residual oils in the heavy oil core after CO_2 flooding.

The unit CO_2 sequestration amount used to eliminate the influence of core size. It was the ratio of the CO_2 mass in the core to the length of the core. Then CO_2 sequestration amount of two cores could be directly compared, as shown in Fig. 10(b). The total unit CO_2 sequestration amount in the low permeability was larger than that in the heavy oil core. The CO_2 sequestration amount of the heavy oil core was mainly in the mesoporous while it was relatively evenly distributed in the macropore and mesoporous for the low permeability light oil core.

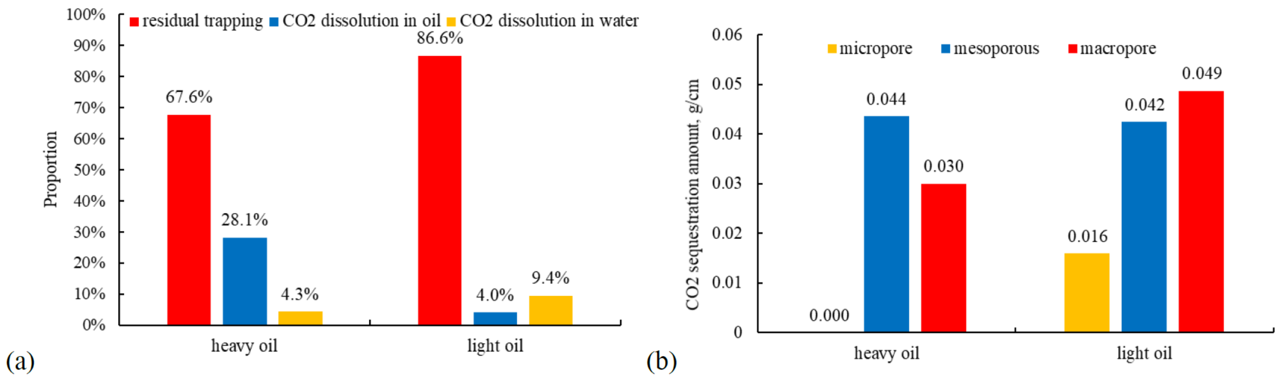

Figure 10—CO_2 sequestration states and amount elevation: (a) Proportion of the three CO_2 sequestration states; (b) Unit CO_2 sequestration amount in each pore sizes

Comparison of CO_2 sequestration potential of two reservoirs at the filed-scale

The CO_2 sequestration potential at the filed-scale could be calculated via the numerical simulation. The residual trapping was closed to the oil recovery. Although the oil recovery of CO_2 flooding in the low permeability reservoir was higher than that in the heavy oil reservoir, the residual trapping proportion in the low permeability reservoir was lower than that in the heavy oil reservoir, as shown in Fig 11. The reason was the reservoir pressure of low permeability reservoir was much higher than that of heavy oil reservoir. It leaded the CO_2 dissolution amounts in oil and water were high. It was different from core-scale results in Fig. 10(a) because CO_2 could full sweep the core but the swept volume was limited at the field-scale. Fig. 11(b) showed the oil recovery of heavy oil reservoir was very sensitive to the temperature. It recommended to firstly conduct thermal recovery to increase the reservoir temperature, and then injection CO2 for enhanced oil recovery and sequestration.

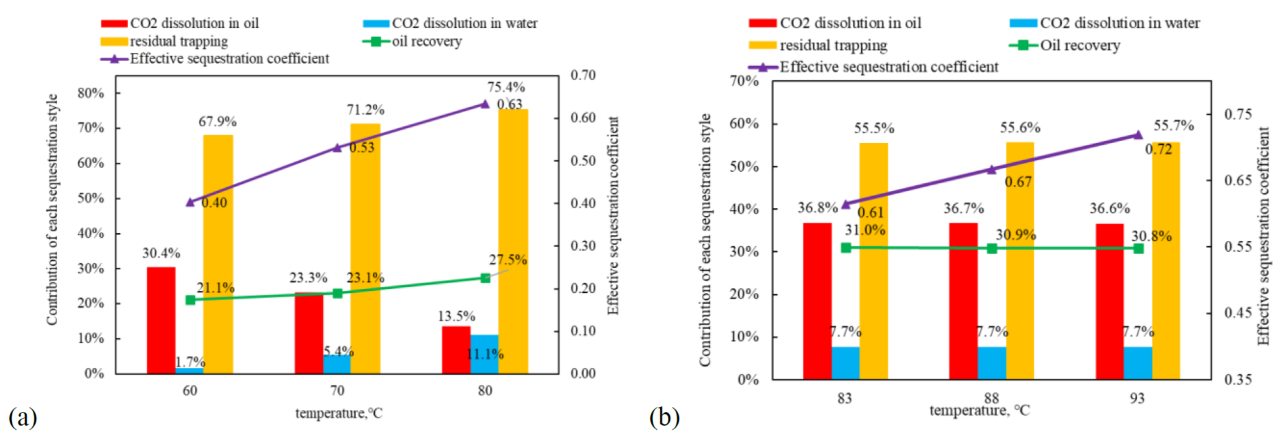

Figure 11—Contribution of each sequestration style and effective sequestration coefficient: (a) heavy oil reservoir; (b) low permeability reservoir

Effective sequestration coefficient was defined as the ratio of the CO_2 sequestration amount underground to the pore volume of the reservoir. Larger effective sequestration coefficient means more space could be used underground for CO_2 sequestration. Since the pressurization stage of CO_2 sequestration also be considered in the simulation, the effective sequestration coefficient of low permeability reservoir was larger than that of heavy oil reservoir. These results indicated the low permeability reservoir may be more suitable for CO_2 sequestration.

Conclusions

This work compared CO_2 enhanced oil recovery and sequestration potential in a heavy oil reservoir and low permeability reservoir via the experimental research and numerical simulation. The conclusions could be drawn as follows:

1. The breakthrough pressure of the mudstone caprock of the low permeability reservoir was 13.4MPa, which was 3.5 times higher than that of the mudstone caprock of the heavy oil reservoir. It indicated the CO_2 sequestration pressure in the low permeability reservoir was higher than that in the heavy oi reservoir.
2. The CO_2 solubility in the heavy oil was 2 times lower than that in the light oil at the same pressure and temperate. They both increased with pressure increasing and decreased with temperate increasing.
3. The CO_2 displacement efficiency of low permeability light oil reservoir was 88.6%, which was much larger than that of heavy oil reservoir of 38.9%. It leaded the CO_2 sequestration amount per unit pore volume in the low permeability light oil reservoir was slightly higher than that in the heavy oil reservoir. The proportion of CO_2 sequestration by capillary trapping in low permeability cores could reach 87% while the CO_2 sequestration by dissolution in the heavy oil reservoir was nearly 30%.
4. Numerical simulation found that in the real reservoir conduction, the residual trapping proportion in the low permeability reservoir was lower than that in the heavy oil reservoir but the effective sequestration coefficient of low permeability reservoir was larger than that of heavy oil reservoir.

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